Potassium permanganate ($KMnO_4$) is a highly reactive inorganic salt that appears as granular dark purple crystals, nearly black in color. Due to its useful chemical properties and the fact that its reactions typically do not produce compounds that are toxic, $KMnO_4$ is employed in a variety of applications, including use as a topical skin disinfectant, an antifungal agent, a fire starter, and a disinfectant for drinking water.

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$$2\ MnO_2 + 4\ KOH + O_2 \rightarrow 2\ K_2MnO_4 + 2\ H_2O$$

Reaction 1

In step 2, the potassium manganate is then converted into potassium permanganate electrochemically according to Reaction 2.

$$2\ K_2MnO_4 + 2\ H_2O \rightarrow 2\ KMnO_4 + 2\ KOH + H_2$$

Reaction 2

In which compound does manganese have a +6 oxidation number?

- A. MnO [1%]
- B. MnO_2 [2%]
- C. K_2MnO_4 [89%]
- D. $KMnO_4$ [5%]

Correct
89% answered correctly

Explanation:

Oxidation number (single atom)	+1	+6	−2
Compound formula	K_2	Mn	O_4
Totals (column)	+2	+6	−8

Zero net charge in the formula, so totals row must sum to zero	(+2) + (+6) + (−8) = 0

Oxidation numbers (oxidation states) are assessed for atoms in a compound as a means of accounting for the gain or loss of electrons by an atom relative to the number of electrons present in a neutral atom of that element. Oxidation numbers may be assessed using Lewis structures; however, for compounds in which all atoms of a given element have equivalent states, values may be alternatively assessed using the compound formula and oxidation state rules. By this second method, atoms with consistent oxidation states (ie, those that rarely vary) are used to determine oxidation states for atoms that have variable states (eg, most metal atoms).

Oxidation numbers differ from the formal charge in that oxidation numbers account for changes caused by oxidation or reduction whereas formal charge accounts for charge distribution among atoms. However, the **sum of all the oxidation states** of every atom in the compound must equal the **net charge** of the compound formula.

Using the **oxidation state rules** for the formula K_2MnO_4 shows that manganese has an oxidation state of +6. Although manganese is a variable metal that has seven possible oxidation states, the oxidation states of oxygen (−2) and potassium (+1) are consistent and can be used to find the value for manganese. Two potassium atoms (each with a state of +1) and four oxygen atoms (each with a state of −2) sum to −6.

$$K_2MnO_4 = 0 \text{ net charge}$$

$$2(+1) + Mn + 4(-2) = 0$$

$$+2 + Mn - 8 = 0$$

Therefore, for the net charge of the formula to equal zero, the manganese atom must have a value of +6.

(Choices A and B) Given that oxygen has an oxidation state of −2, manganese must have a +2 and +4 oxidation state in MnO and MnO_2, respectively for the net charge to equal zero.

(Choice D) The oxidation state of manganese in $KMnO_4$ is +7 rather than +6.

Educational objective:
Using oxidation state rules, the oxidation number of each atom in a compound formula may be determined on the basis that the sum of the oxidation states of each atom must equal the net charge of the compound or ion.

Time spent: 0 sec

Subject:
General Chemistry

QID: 401592

System:
Atomic Theory and Chemical Composition

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$$2\ MnO_2 + 4\ KOH + O_2 \rightarrow 2\ K_2MnO_4 + 2\ H_2O$$

Reaction 1

In step 2, the potassium manganate is then converted into potassium permanganate electrochemically according to Reaction 2.

$$2\ K_2MnO_4 + 2\ H_2O \rightarrow 2\ KMnO_4 + 2\ KOH + H_2$$

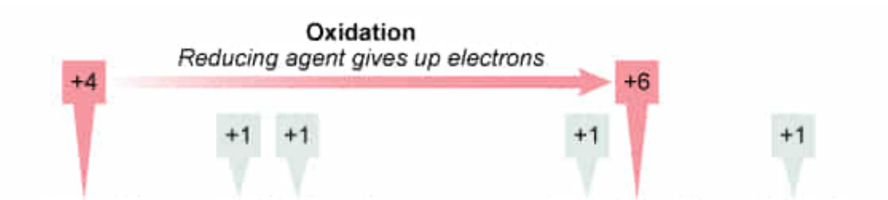
Reaction 2

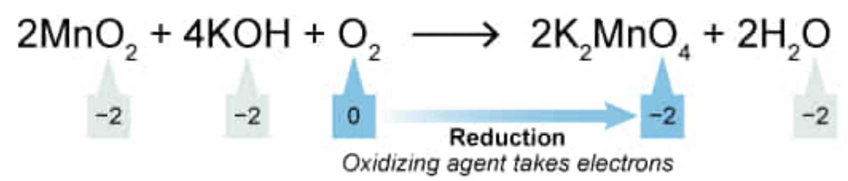
In Reaction 1 shown in the passage, the oxidizing agent is:

- A. K_2MnO_4, because potassium loses electrons. [3%]
- B. MnO_2, because manganese gains electrons. [14%]
- C. KOH, because hydrogen loses electrons. [8%]
- ✔ D. O_2, because oxygen gains electrons. [73%]

Correct
74% answered correctly

Explanation:





O_2 oxidizes MnO_2 by taking up electrons, but O_2 gets reduced in the process

Oxidation-reduction (redox) changes occurring within a reaction can be evaluated by assessing the oxidation number of each atom in the product and reactant species. By **comparing the oxidation number** for a given element in the reactants with the oxidation number of that same element in the products, a loss of electrons (oxidation) or a gain of electrons (reduction) can be identified. Elements that have oxidation numbers that do not change are not involved in redox processes.

Oxygen (O_2) is among the simplest and most readily available of the common oxidizing agents. For the oxidation of MnO_2 (Reaction 1), O_2 acts as the oxidizing agent because it **causes oxidation** (ie, the loss of electrons) in MnO_2 by taking two electrons from the manganese atom. In the process, O_2 itself **gets reduced** (ie, the oxygen atoms gain electrons). This interaction is reflected in the oxidation numbers. Manganese goes from an oxidation state of +4 to an oxidation state of +6 (a loss of two electrons) whereas the oxygen atoms in O_2 go from an oxidation state of 0 to a state of −2 (a gain of two electrons). The states of the other atoms in the reaction do not change and are not redox active.

(Choices A and C) Potassium and hydrogen in K_2MnO_4 and KOH, respectively, are not oxidized or reduced, and therefore are not oxidizing or reducing agents. K_2MnO_4 is the product of the oxidation reaction rather than the oxidizing agent.

(Choice B) MnO_2 is the reducing agent. MnO_2 gets oxidized (gives up electrons) and causes reduction in O_2.

Educational objective:
By comparing the oxidation number for a given element in the reactants with the oxidation number of that same element in the products, oxidation and reduction processes can be identified. Oxidizing agents cause oxidation in other atoms while the agent itself gets reduced in the process.

Time spent:0 sec

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System:
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Reaction 1

In step 2, the potassium manganate is then converted into potassium permanganate electrochemically according to Reaction 2.

$$2\ K_2MnO_4 + 2\ H_2O \rightarrow 2\ KMnO_4 + 2\ KOH + H_2$$

Reaction 2

As an alternative to the industrial preparation outlined by Reactions 1 and 2, potassium permanganate can be prepared in the lab by each of the reactions shown below. All of these can be classified as a disproportionation reaction EXCEPT:

I. $2\ K_2MnO_4 + Cl_2 \rightarrow 2\ KMnO_4 + 2\ KCl$

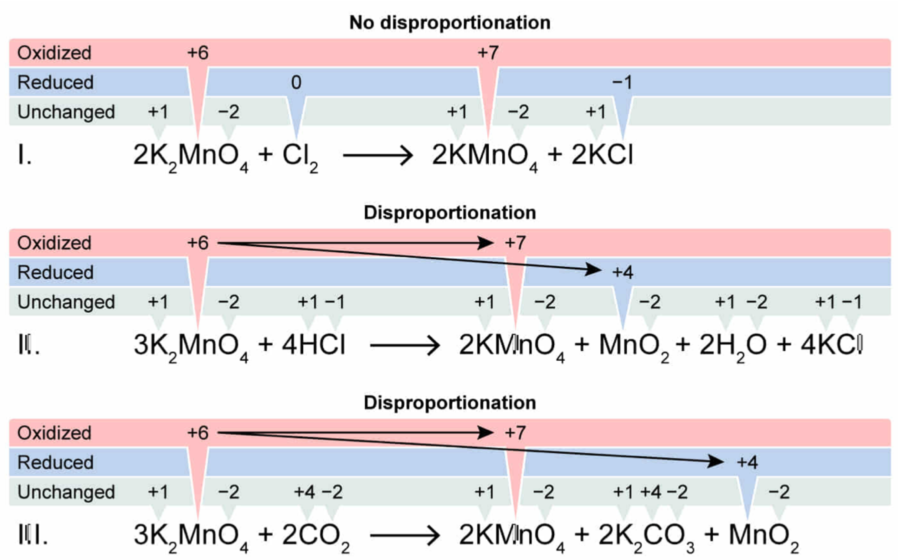
II. $3\ K_2MnO_4 + 4\ HCl \rightarrow 2\ KMnO_4 + MnO_2 + 2\ H_2O + 4\ KCl$

III. $3\ K_2MnO_4 + 2\ CO_2 \rightarrow 2\ KMnO_4 + 2\ K_2CO_3 + MnO_2$

- A. I only [52%]
- B. III only [16%]
- C. I and II only [12%]
- D. II and III only [18%]

Correct
52% answered correctly

Explanation:



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A **disproportionation reaction** is a special case of an oxidation-reduction (redox) reaction in which both the **oxidation** and the **reduction** occur to atoms of the **same element**. Disproportionation reactions can be easily identified by assigning oxidation numbers and then comparing the oxidation number of the given element in the reactants with the oxidation number of that same element in the products.

For elements undergoing oxidation, the oxidation numbers increase whereas oxidation numbers decrease for elements undergoing reduction. Oxidation numbers that do not change for elements that are not involved in redox processes. If a disproportionation reaction occurs, then the same element (at the same initial oxidation state) will provide both the atoms being reduced as well as the atoms being oxidized.

In Reaction I, two elements are involved in the redox process. Manganese is oxidized from a state of +6 to a state of +7, and chlorine is reduced from its elemental

state to a state of −1. Therefore, no disproportionation occurs in this reaction because the oxidation and reduction involve different elements.

(Numbers II and III) In Reactions II and III, both the oxidation and the reduction occur to the *same element* (manganese). All atoms of manganese atoms start at an oxidation state of +6, but some of these atoms are further *oxidized* to a state of +7 whereas other manganese atoms are *reduced* to a state of +4. Therefore, disproportionation occurs in these reactions.

Educational objective:
A disproportionation reaction is a special case of an oxidation-reduction reaction in which both the oxidation and the reduction occur to atoms of the same element. Disproportionation reaction can be identified by comparing the oxidation number of a given element in the reactants with the oxidation number of that same element in the products.

Time spent:0 sec

Subject:
General Chemistry

QID:401594

System:
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$$2\ MnO_2 + 4\ KOH + O_2 \rightarrow 2\ K_2MnO_4 + 2\ H_2O$$

Reaction 1

In step 2, the potassium manganate is then converted into potassium permanganate electrochemically according to Reaction 2.

$$2\ K_2MnO_4 + 2\ H_2O \rightarrow 2\ KMnO_4 + 2\ KOH + H_2$$

Reaction 2

What volume of H_2 gas will be generated in Reaction 2 described in the passage if 10.00 L of a 5.00 M solution of K_2MnO_4 react at standard temperature and pressure (STP)?

- A. 1.12 L [8%]
- B. 22.4 L [19%]
- C. 560 L [62%]
- D. 1120 L [9%]

Correct
62% answered correctly

Explanation:

$$2K_2MnO_4 + 2H_2O \longrightarrow 2KMnO_4 + 2KOH + H_2$$

Volume reacting		K_2MnO_4 solution molarity		Mole ratio of reaction		Molar volume of gas at STP		Volume produced
$10.0\ \cancel{L}$ solution	×	$\frac{5.00\ \cancel{mol}\ K_2MnO_4}{1\ \cancel{L}\ \text{solution}}$	×	$\frac{1\ \cancel{mol}\ H_2}{2\ \cancel{mol}\ K_2MnO_4}$	×	$\frac{22.4\ L\ \text{gas}}{1\ \cancel{mol}\ \text{gas}}$	=	$560\ L\ H_2$

The **molarity** of a solution is the number of moles of solute present in 1 L of solution. The number of moles of reactant can be related to the number of moles of product formed in a reaction by the stoichiometric mole ratios obtained from the balanced reaction equation. If the reaction product is a gas, the volume the gas will occupy depends on both temperature and pressure.

The volume of a gas is often stated under conditions of **standard temperature and pressure (STP)**, which are defined as 0°C (273 K) and 1 atm pressure. Under STP, 1.00 mole of gas occupies a volume of 22.4 L. This molar volume (which is only valid at STP) is often just used as a known constant, but this value can be derived using the ideal gas law as follows:

$$\mathrm{PV} = \mathrm{nRT}$$

$$\frac{V}{n} = \frac{\mathrm{RT}}{P} = \frac{\left(0.0821\ \frac{\mathrm{L} \cdot \cancel{\mathrm{atm}}}{\mathrm{mol} \cdot \cancel{\mathrm{K}}}\right)(273\ \cancel{\mathrm{K}})}{(1.00\ \cancel{\mathrm{atm}})} = 22.4\ \mathrm{L/mol}$$

For Reaction 2, the number of moles of K_2MnO_4 present in 10.0 L of solution can be calculated by multiplying this volume by the molar concentration (molarity) of the solution.

$$10.0\ \cancel{\mathrm{L\ solution}} \times \frac{5.00\ \mathrm{mol\ K_2MnO_4}}{\cancel{\mathrm{L\ solution}}} = 50.0\ \mathrm{mol\ K_2MnO_4}$$

The mole ratios obtained from the balanced equation for

Reaction 2 show that 1 mole of H_2 is produced for every 2 moles of K_2MnO_4 that react. Applying this ratio, the number of moles of H_2 produced can be determined:

$$50.0\ \cancel{\text{mol } K_2MnO_4} \times \frac{1 \text{ mol } H_2}{2\ \cancel{\text{mol } K_2MnO_4}} = 25.0 \text{ mol } H_2$$

At STP, the **molar volume** occupied by the H_2 gas produced in the reaction is given by:

$$25.0\ \cancel{\text{mol}}H_2 \times \frac{22.4 \text{ L}}{1\ \cancel{\text{mol}}} = 560 \text{ L } H_2$$

(Choice A) If the number of moles of H_2 is divided rather than multiplied by 22.4 L/mol, a result of 1.12 L is obtained.

(Choice B) If the reacting solution volume is divided instead of multiplied by the K_2MnO_4 solution molarity, a result of 22.4 L is obtained.

(Choice D) If a 1:1 (H_2:K_2MnO_4) mole ratio is used rather than a 1:2 mole ratio, a value of 1120 L results.

Educational objective:
Given the volume of a solution, its molarity, and a balanced reaction, mole ratios can be determined that relate the moles of a reactant present in the solution to the moles of a product produced from the solution. For gaseous products, the number of moles produced can be

related to the molar volume of 22.4 L/mol, the volume 1 mole of gas occupies at STP.

Time spent:0 sec

Subject:
General Chemistry

QID:401595
System:
Atomic Theory and Chemical Composition

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$$2\ MnO_2 + 4\ KOH + O_2 \rightarrow 2\ K_2MnO_4 + 2\ H_2O$$

Reaction 1

In step 2, the potassium manganate is then converted into potassium permanganate electrochemically according to Reaction 2.

$$2\ K_2MnO_4 + 2\ H_2O \rightarrow 2\ KMnO_4 + 2\ KOH + H_2$$

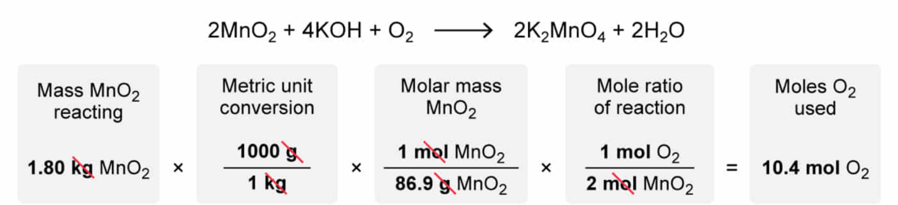
Reaction 2

How many moles of O_2 are consumed if 1.8 kg of MnO_2 are used in Reaction 1? (Note: The molar mass of MnO_2 is 86.9 g/mol.)

- A. 0.010 mol [5%]
- B. 10.4 mol [78%]
- C. 20.7 mol [11%]
- D. 41.4 mol [4%]

Correct
79% answered correctly

Explanation:



The molar mass of a compound is the amount of mass contained in 1 mole of the compound (ie, the grams per mole) and can be used to convert the **mass** of a compound to the number of **moles**. The reactants and/or products of a particular reaction are related to one another by the **mole ratios** from the balanced reaction equation.

For Reaction 1, the number of moles of MnO_2 present in the reaction can be calculated by dividing the given mass of MnO_2 (1.8 kg = 1800 g) by its molar mass:

$$1800\ \cancel{\text{g MnO}_2} \times \frac{1\ \text{mol MnO}_2}{86.9\ \cancel{\text{g MnO}_2}} = 20.7\ \text{mol MnO}_2$$

Reaction 1 is balanced and the mole ratios show that 1 mole of O_2 is consumed for every 2 moles of MnO_2 that react. Applying this ratio, 20.7 moles of MnO_2 consumes half as many moles of O_2:

$$20.7\ \cancel{\text{mol MnO}_2} \times \frac{1\ \text{mol O}_2}{2\ \cancel{\text{mol MnO}_2}} = 10.4\ \text{molO}_2$$

(Choice A) If the mass of MnO_2 is not converted from kilograms to grams, a value of 0.010 mole (with incorrect units) is obtained.

(Choice C) If a 1:1 mole ratio between MnO_2 and O_2 is used rather than a 2:1 ratio, 20.7 moles is the result.

(Choice D) If a 2:1 mole ratio between MnO_2 and O_2 is inverted, resulting in multiplication instead of division by 2 during the calculation, 41.4 moles is obtained.

Educational objective:

Given the mass of a compound sample, the number of moles present in the sample can be determined using its molar mass (the amount of mass present in 1 mole of a substance). Relating a known number of moles of a reactant to a balanced reaction equation, mole ratios can be applied to determine the number of moles of one chemical species relative to the number of moles of another chemical species in the reaction.

Time spent: 0 sec

QID:401596

Subject:
General Chemistry

System:
Atomic Theory and Chemical Composition

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Reaction 1

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$$2\ K_2MnO_4 + 2\ H_2O \rightarrow 2\ KMnO_4 + 2\ KOH + H_2$$

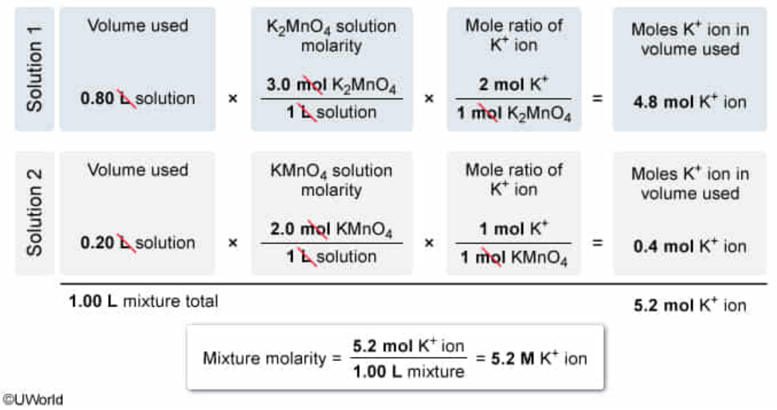
Reaction 2

Suppose 0.80 L of 3.0 M aqueous K_2MnO_4 is combined with 0.20 L of 2.0 M aqueous $KMnO_4$ to make a solution. What is the concentration of potassium ions in the solution?

- A. 2.8 M [24%]
- B. 4.0 M [10%]
- C. 5.2 M [60%]
- D. 6.0 M [4%]

Correct
60% answered correctly

Explanation:



When a given volume is dispensed from a stock solution, a specific number of moles of solute are transferred with the sample. The number of moles in the sample can be calculated by multiplying the molarity (mol/L) of the stock solution by the volume (L) of the sample. When two samples from different stock solutions are mixed, the final molar **concentration of the sample mixture** is determined by the total number of **moles transferred** (of a given species) divided by the **total volume** of the resulting mixture.

For the K_2MnO_4 stock solution, a 0.80 L sample transfers 4.8 mol of K^+ ions, and for the $KMnO_4$ stock solution, a 0.20 L sample transfers 0.4 mol of K^+ ions. When the samples are mixed, a total of 5.2 mol of K^+ ions (4.8 mol + 0.4 mol) are present in a mixture volume of 0.80 L + 0.20 L = 1.00 L. Therefore, the concentration of K^+ ions resulting from mixing the samples is 5.2 mol/L.

(Choice A) If a 1:1 mole ratio is used instead of the 2:1 mole ratio of K^+ ions to K_2MnO_4 present in the potassium manganate solution, 2.8 M is obtained from the calculation.

(Choice B) The average K^+ ion concentration in the two stock solutions is 4.0 M. Because unequal volumes from the two stock solutions are mixed, the average concentration of the stock solutions is not equivalent to the concentration of the mixture.

(Choice D) The K^+ ion concentration in the K_2MnO_4 stock solution is 6.0 M. The 2:1 mole ratio between K^+ ions and K_2MnO_4 makes the K^+ ion concentration double that of the K_2MnO_4 concentration.

Educational objective:
The number of moles transferred in a sample taken from a stock solution can be calculated by multiplying the molarity of the stock solution by the sample volume. When two samples from different stock solutions are mixed, the molarity of the sample mixture equals the total number of moles transferred divided by the total mixture volume.

Time spent:0 sec	**QID**:401597
Subject:	**System:**
General Chemistry	Atomic Theory and Chemical Composition